

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Losing Screen Efficiency: Blinding, Pegging and Carryover

Field notes from Jonathan Gilmour. Independent controls engineer, PLC/HMI/SCADA, Bay Area and Northern California.

A screen that used to make spec quietly stops making it, and the calls I get run the same way every time: "the SCADA's fine, but the product's dirty." Screening is a mechanical separation dressed up with a little instrumentation, so the honest starting point is that most efficiency losses are not a setpoint problem. They're moisture, worn cloth, a stroke that's drifted, or discs wrapped in film. The controls help you see the loss and hold a steady feed, and once in a while the controls are the fix. But if you go chasing a HMI number before you've put eyes and a hand on the deck, you'll burn a shift. This guide is for the maintenance lead or screen tech standing at a blinded deck at 2am on an aggregate crushing plant or a materials recovery facility (MRF, the plant that sorts single-stream recycling), trying to decide what to touch first.

Let me fix the vocabulary first, because the crew and the plant manager usually mean three different things by the same word, and that alone sends people to the wrong repair.

Say what you actually mean: blinding vs pegging vs carryover

Blinding is fine, damp or sticky material coating the cloth and bridging across the openings. Think clay, crusher fines with moisture in them, dirt, dust that's picked up humidity, sludge on a MRF fines screen. It builds up as a skin over the wire and effectively shrinks or seals the aperture (the opening in the cloth). Blinding involves material much finer than the opening — it's a coating problem, and it usually shows up in patches or across a whole zone of the deck.

Pegging (also called plugging) is near-size particles physically wedged into the aperture, one stone per hole. A near-size particle is one just a hair bigger than the opening, so it enters partway and jams. Pegging is a geometry problem, not a moisture problem, and you'll see it as individual stuck stones you could knock out with a bar, often worst where the material bed is thinnest and the near-size fraction is heavy.

Carryover and contamination is the result, not a cause. Carryover is undersize material riding over the end of the deck and reporting to the wrong pile because blinding or pegging closed off the openings it should have dropped through. On aggregate you'll hear it as "oversize in my product" or a

dirty base rock. On a MRF disc screen you'll hear it as "fiber in the containers" or "containers in the fiber" — misplaced material either way. Carryover is your symptom. Blinding and pegging are the two mechanisms behind it, and they get fixed differently, which is exactly why the words matter.

One more distinction that saves arguments: blinding gets *worse* the longer the machine runs and often clears itself when the feed dries out or you clean the deck; pegging is stable and physical and won't clear until something knocks the stones loose or you change the media. If the deck cleans up after a rain stops or after you cut damp feed, you were blinding. If it's the same after a week of dry weather, you're pegging.

| Ten-minute triage

Before any teardown, lock out and tag out (lockout/tagout, LOTO) the machine — see the safety section, this is not optional before you're over a deck. Then:

1. **Look at the discharge (0-2 min).** Where does undersize start riding over instead of dropping through? That point on the deck — feed end, middle, discharge end — tells you a lot. Blinding tends to start at the feed end where the wet load hits; pegging spreads where the bed thins out toward discharge.
2. **Grab the carryover (2-4 min).** Scoop a handful of what's coming off the wrong end. Is it undersize that should have passed (efficiency loss) or genuinely oversize (that's correct separation, look elsewhere)? Undersize in the overs pile confirms you're losing efficiency, not mis-set.
3. **Read the cloth, LOTO'd, hand on the deck (4-7 min).** Coating you can smear with a gloved thumb is blinding. Individual stones stuck one-per-hole are pegging. Holes, worn wires, or a slack section are media wear. A whole zone sealed flat is heavy blinding.
4. **Feel and hear the motion (7-9 min).** Restart if safe, or watch from the guarded side. Is the stroke lively and throwing material, or lazy? Is one corner dead? A tennis ball or a stroke card (the peel-and-stick amplitude card) tells you fast. A bearing rumble or a knock is its own problem.
5. **Check the feed (9-10 min).** Is it centered, steady, and within rated tonnage, or surging and off to one side? Damp and lumpy, or dry and running? Overfed decks blind and peg both.

That flow points you at feed, media, or motion in ten minutes. The rest of this guide works each one in order — safest and most-likely first.

| Cause 1: Feed — start here, it's the cheapest fix

More screening problems are born at the feed than anywhere else, and feed is usually the thing you can change today without buying parts.

Moisture. This is the number one driver of blinding. Fines that would fall freely when dry turn into a paste that plasters the cloth when they carry a few percent of surface moisture. There's a nasty middle band — material that's damp enough to stick but not wet enough to wash through — and screening in that band is miserable. If your blinding tracks the weather, tracks a wash step upstream, or is worst on

the first pile of the morning off a wet stockpile, moisture is your answer. Log the material moisture if you can sample it, or at minimum log "wet/damp/dry" and the weather each time it blinds. Real fixes live upstream: dewater or age the stockpile, feed off a drier part of the pile, add a heater or wash where the process allows, or accept that a genuinely wet feed wants non-blinding media (below).

Clay and plastic fines. Clay is worse than plain water because it's sticky when barely damp and it bridges apertures like mortar. If you're screening pit-run with clay seams or a MRF fines line with wet paper and food residue, expect blinding that a dry day won't clear. This is a media and sometimes a wash-plant conversation, not a screen-motion one.

Overloading. A bed that's too deep can't stratify — the fines never work their way down to the cloth to present themselves to an opening, so they ride over as carryover even with a perfectly clean deck. Overloading also packs near-size into apertures and drives pegging. Check tonnage against the machine's rated capacity for your cut and material (check against your equipment's spec — capacity charts are per square foot of deck, per aperture, and derate hard for moisture and fines). If you're feeding a screen sized for 200 tph at 300 because the crusher's running hot, the screen is not the problem.

Off-center or surging feed. Material dumped to one side of the deck overloads that side and starves the other; you get carryover on the heavy side and wasted open area on the light side. Surging feed — slugs coming off a stop-start conveyor or a hopper that bridges and lets go — does the same thing in time instead of space, and it's brutal on separation because the screen never reaches steady state. Look at the feed spread across the full deck width. A worn or misaligned feed chute, a dribble conveyor running empty then slugging, or a bin that bridges are the usual culprits.

Feed-rate control. This is the first real controls lever. A screen fed at a steady, correct rate outperforms the same screen fed in surges every time, even if the average tonnage is identical. If you've got a variable-frequency drive (VFD, the drive that varies motor speed) on the feeder and a level or weigh signal, hold the feed steady and choke-feed the hopper (keep it flooded so material fills the feeder and it never runs empty) so the feeder — not the upstream surge — sets the rate. On a lot of plants this single change recovers more efficiency than anything you'll do to the deck. More on the controls in its own section.

■ Cause 2: Media — the cloth itself

If feed checks out, the cloth is next, and it's the most commonly wrong thing on old machines because it wears and because somebody in the past grabbed whatever was on the shelf.

Wrong aperture or wire diameter. Confirm the installed opening actually matches the cut you're trying to make, and that nobody swapped in a heavier wire to make cloth last longer. Heavier wire means smaller open area for the same nominal opening, and less open area means less capacity and more carryover. Log the actual aperture and wire size off the cloth or the packing slip, not what the drawing says.

Worn or holed cloth. Worn wires open up the aperture and let oversize through (contamination of your product); a hole does the same locally and dramatically. Wear concentrates where the feed lands and along the discharge lip. Run a hand — LOTO'd — along the high-wear zones and look for thinned wires, elongated openings, and pumping (cloth flexing loose because tension's gone). A screen "losing efficiency" is sometimes a screen that's now passing oversize because the cloth is shot.

Open area. Open area is the percentage of the cloth that's actually holes. Two cloths with the same nominal aperture can differ a lot in open area depending on wire and weave. Low open area chokes capacity and promotes carryover; that's the hidden cost of the heavy-wire swap above. If you're chronically short on capacity, more open area — thinner wire, or a more open weave, or an engineered polyurethane or rubber panel made in a higher-open-area version — may buy back the throughput without changing the cut. Be honest about the trade: most polyurethane and rubber panels give up some open area to gain wear life and anti-blinding, so confirm the panel's rated open area against the wire cloth it's replacing before you count on it for capacity.

Tensioning. Tension is the quiet killer on tensioned-cloth machines. A cloth that's lost tension doesn't get the sharp local acceleration that throws blinding material clear — it goes slack and dead, blinds fast, and wears in a hurry from pumping against the deck bearers. Retension to the machine's procedure, check the crown bar rubber and the side tension rails, and log the date. A lot of "the media's junk" calls are really "the media's loose."

Non-blinding media options. This is the legitimate mechanical fix for chronic blinding and pegging, and it's often the right answer instead of chasing setpoints. Self-cleaning wire (wires that vibrate independently, like piano-wire or ondap/zigzag weaves), polyurethane or rubber modular panels with tapered or slotted apertures that shed near-size, and rubber-ball or slider decks (loose rubber balls under the cloth that bounce and knock the underside clear) all exist specifically to beat blinding and pegging. Tapered apertures — wider at the bottom than the top — let a near-size particle that enters keep going instead of wedging, which directly attacks pegging. If a deck blinds or pegs no matter what you do to feed and motion, the media is telling you it's wrong for the material. That's a purchase, not a knob, and it's an honest answer.

■ Cause 3: Machine motion — stroke, speed, timing, angle

Vibrating screens separate by throwing material up and forward so it stratifies and the fines present to the cloth. Get the motion wrong and even perfect cloth blinds and carries over. Warning: several checks here involve running the machine and getting near rotating counterweights and an energized drive. Keep guards on for running checks, stand clear of the counterweight arc, and do any hands-on adjustment only in a full zero-energy LOTO state — see safety.

Stroke and amplitude. Stroke (the total throw, peak to peak) sets how hard material is thrown. Too little stroke and the bed doesn't loosen — fines stay trapped in the bed, blinding material isn't shaken clear, and you carry over. Too much and you can throw near-size straight over the deck or beat the

machine to death. Measure it with a stroke card stuck to the side plate while running; compare to the machine's target (check against your equipment's spec — it's typically stated as a stroke range for the material and cut). Log the reading at both ends and both sides.

Speed and frequency. Speed (rpm, or the operating frequency) works with stroke to set the g-force and the throw. Low speed and low g means a lazy bed and blinding; too high can float the bed and reduce contact. Aggregate screens commonly run in a broad range depending on size and duty (again, your spec sets the target — don't invent one). On a VFD-driven or electric-vibrator machine you can actually trim this; on a fixed exciter you're stuck with the sheave and the counterweights.

Timing and counterweights. On dual-shaft or exciter machines the counterweight setting and the timing between shafts set the stroke angle and whether you get a straight-line, circular, or elliptical throw. If somebody changed counterweights, or a timing gear or belt slipped, the throw angle is wrong and material either dwells too long (blinds, pegs) or races over (carryover of undersize). Confirm the counterweight setting against the machine's chart and log it. This is a real and under-checked cause after any exciter service.

Deck angle. Deck slope sets how fast material travels and therefore how long it dwells over the openings. Too steep and material sprints to discharge without time to stratify and pass — classic carryover. Too shallow and it dwells, builds bed depth, and blinds. If someone shimmed the machine or the base settled, the angle drifted. Check it with a level against the machine's spec angle and log it.

Out-of-balance and bearings. A dead corner, an orbit that's gone from circular to a flat ellipse, or a new rumble means a mechanical fault — cracked side plate, loose counterweight, failing vibrator bearing, broken spring, or debris packed in a spring mount. This both wrecks separation and destroys the machine if you run it. An orbit sheet (draw the machine's motion at each corner using a motion card) is the standard check; a bearing that's running hot or rumbling gets pulled, not tuned. This is where vibration monitoring earns its keep, below.

■ Cause 4: Disc and star screen specifics (MRF)

Disc screens and star screens are a different animal. Instead of a vibrating cloth, they use rows of rotating shafts carrying interlocking discs or rubber stars; the gaps between discs are the "aperture," and the rotation carries oversize (fiber, cardboard) across the top while smalls and containers drop through the gaps. Same three failure words, different mechanics.

Disc spacing. The gap between discs is your cut point, and it's set by disc and spacer geometry. Wrong spacing sends the wrong stream down — too tight and fiber-appropriate material won't carry, too open and containers ride over with the fiber (carryover). Confirm spacing against the screen's setup for the cut you want, and log it.

Wrapping. This is the MRF version of blinding, and it's the number one disc-screen killer: film plastic, plastic bags, tape, stretch wrap, and long stringy material wrap around the shafts and discs, close the gaps, and turn a sorting screen into a conveyor that carries everything over. You'll see it as fiber

quality collapsing and containers reporting to the fiber line. The fix is scheduled de-wrapping (cutting the wrap off the shafts) — LOTO first, every time, because reaching into a disc screen that isn't zero-energy is how people lose fingers and hands. Log wrap severity and de-wrap intervals; if you're de-wrapping every shift, that's a front-end sort or a screen-selection problem upstream, not just housekeeping.

Worn discs and stars. Rubber stars and disc rubber wear, and worn profiles lose their grip and their gap geometry, so the cut drifts and carryover climbs. Look for rounded, cracked, or missing discs and log the worst rows.

Shaft speed. Disc-screen shaft speed sets agitation and throughput; too slow and material blinds the bed and mis-sorts, too fast and it launches material over. On VFD-driven screens this is a real adjustable, and it's one of the few places on a MRF screen where a setpoint genuinely moves separation. Trim it against the screen's guidance and the observed carryover, and log before/after.

| How to actually measure efficiency

You cannot tune what you can't measure, and "it looks dirty" isn't a number. Two field methods:

Grab a gradation. Take a representative grab of the product and run a sieve analysis (a gradation, the stack of test sieves that tells you the size distribution). The question is simple: how much undersize is in the overs (that's screening inefficiency), or how much oversize is in the product (that's contamination from worn or holed cloth). Screen efficiency is commonly figured as the fraction of the available undersize that actually reported to the undersize product; you don't need the textbook formula at 2am, you need the trend. Sample the same way each time — same spot, same size grab, same sieves — and log it, because the absolute number matters less than whether it's getting better or worse after each change you make.

Watch the carryover point. On the running deck (guarded, from the safe side) find where undersize stops dropping through and starts riding over. That location moving toward the feed end means the effective screening area is shrinking — blinding or pegging progressing, or bed too deep. Watching that point shift is the fastest live feedback you get, and it's how you tell whether the last thing you changed helped. Note the position each time.

Make one change at a time and re-measure. If you change feed rate, media tension, and deck angle all at once, you've learned nothing and you can't back out the one that hurt.

| The controls angle — and where controls aren't the answer

Here's the honest map of what the PLC/SCADA can and can't do for a blinding screen.

Feed-rate and level control — real, and usually worth doing. Steady feed is the single biggest controls contribution to screening. A VFD on the feeder holding a constant rate off a choke-fed, level-controlled hopper takes the surging out and lets the screen run at its best point. On a MRF, a level or

belt-loading signal that trims the feed metering does the same for the sort screens. If your feed is surging and you have the drives to fix it, that's the highest-value control work on the whole plant.

Screen-motion and bearing-vibration monitoring — real, mostly for protection and early warning. Vibration sensors on the screen body and on the vibrator/exciter bearings catch the out-of-balance, the failing bearing, the cracked side plate, and the lost counterweight before they turn into a scrap machine or a shift-ending failure. Some setups trend the stroke and orbit continuously. This won't unblind a deck, but it tells you when the motion has changed — which is a genuine efficiency cause — and it keeps you from running a wounded machine into the ground. Worth having, especially on a hard-to-inspect exciter.

Where the honest answer is mechanical or media, not a setpoint. Blinding from moisture and clay, pegging from near-size, worn or holed cloth, lost tension, wrong open area, wrapped disc shafts — none of these have a knob. I've been called in to "tune the controls" on screens where the real fix was a load of non-blinding polyurethane panels or a de-wrap schedule, and no amount of setpoint work would have touched it. The controls can hold the feed steady and warn you when the machine's motion drifts. The controls cannot make wet clay fall through a blinded cloth. If you've steadied the feed and the deck still blinds or pegs, stop looking at the HMI and change the media or the feed material. That's not a controls failure, it's the process telling you where the problem actually lives.

| Safety — read before you go near the deck

Screens are high-energy machines in dusty, sometimes confined plants. The hazards here are specific, not boilerplate.

LOTO before you clear a deck or touch media. Any time you're going to be over the deck, knocking out pegged stones, smearing a gloved thumb on the cloth to check for blinding, retensioning, or reaching into a disc screen to cut wrap, the machine must be at a full zero-energy state: locked and tagged out, verified de-energized with a try-to-start, and stopped moving. A vibrating screen and a disc screen both store energy in springs, in the rotating exciter/counterweights, and in loaded material that can shift. People are hurt and killed reaching into disc screens and vibrating decks that "were about to stop" or that a coworker bumped back on. Zero-energy verified, or you don't put a hand in.

Guarding stays on for running checks. Stroke cards, orbit reads, and carryover watching all happen with guards in place, from the designated safe side, clear of the counterweight arc and the drive belts. Never read a stroke card by reaching past a guard on a running machine. If you can't see it safely from outside the guard, shut down and LOTO.

Crushing-plant dust and lifting. Aggregate screening throws respirable crystalline silica dust — the fine stuff from crushed rock that scars your lungs and doesn't announce itself. Wear the respiratory protection the plant's dust program calls for, keep the dust suppression running, and don't stand in the plume. Screen cloth panels, tension rails, and disc-screen shafts are heavy and awkward; use proper lifting gear and a second set of hands, because a media change is where backs and fingers get hurt.

Confined spaces around the plant. Screen underpans, chute boxes, surge tunnels under stockpiles, and bin/silo work near the screen line can be permit-required confined spaces with atmospheric and engulfment hazards. The atmosphere in these spaces can be oxygen-deficient, and where organics or standing water collect — MRF baler pits and pick-line tunnels especially — decomposition can build up carbon monoxide (CO), hydrogen sulfide (H₂S), and methane. Test the atmosphere before entry (oxygen level, flammable gas as a percentage of the lower explosive limit (LEL), CO, and H₂S) and wear a bump-tested personal 4-gas monitor the whole time you're inside. Never enter a bin or silo under a live draw — flowing material engulfs and buries a person in seconds — and never enter any of these without the permit, atmospheric testing, and the entry procedure. On a MRF, hold the same confined-space discipline around baler pits and pick-line tunnels.

Symptom -> likely cause -> check (and log)

Symptom	Likely cause	Check (LOTO first for hands-on) and what to log
Undersize riding over the discharge end (carryover)	Bed too deep / overload, or deck too steep, or blinded/pegged cloth	Log tonnage vs rated capacity; log carryover start position on deck; inspect cloth for coating vs stuck stones
Cloth coated, smears under a gloved thumb, worst at feed end	Blinding from moisture/clay fines	Log material wet/damp/dry and weather; sample moisture if possible; check for wash step or wet stockpile upstream
Individual stones wedged one-per-hole, stable over days	Pegging by near-size particles	Log which aperture and where on deck; compare feed gradation near-size fraction; consider tapered/non-blinding media
Oversize showing up in the product	Worn or holed cloth, or lost tension	Inspect high-wear zones for thinned wires, elongated openings, holes; check and log tension; check crown/side rubber
Carryover on one side, wasted area on the other	Off-center feed	Watch feed spread across full width; check chute alignment; log which side
Separation swings shift to shift with no cloth change	Surging feed	Watch for slugging conveyor/bridging hopper; log feed steadiness; steady with feeder VFD/level control
Lazy bed, fines not stratifying, slow travel	Low stroke or low speed, or lost tension	Stroke card both ends/sides running (guarded); log reading vs spec; check tension

Symptom	Likely cause	Check (LOTO first for hands-on) and what to log
Material racing over without passing	Deck too steep, or stroke too high, or timing/counterweight wrong	Level the deck vs spec angle; log angle; check counterweight/timing after any exciter service
Dead corner, new rumble, orbit gone elliptical	Out-of-balance, bearing, cracked plate, broken spring	Orbit/motion card each corner (guarded); check bearing temp; pull and inspect — do not keep running; log which corner
MRF: fiber quality collapses, containers in fiber	Disc shafts wrapped with film/tape (blinding equivalent)	LOTO, inspect shafts for wrap; log wrap severity and de-wrap interval; if every shift, look upstream
MRF: cut point drifting, more carryover over time	Worn discs/stars or wrong spacing	Inspect for rounded/cracked/missing discs; confirm spacing vs setup; log worst rows
MRF: mis-sort at steady feed	Shaft speed off	Trim shaft speed (VFD) vs guidance and observed carryover; log before/after gradation

Closing note

Work it in order — feed, media, motion, then the disc specifics — measure with a real grab and the carryover point, and change one thing at a time with a line in the log every time. The controls belong in this story mostly to hold your feed steady and to warn you when the machine's motion has drifted; they are not going to make wet clay fall through a blinded cloth, and it's fair to your plant to say so out loud rather than tune around a mechanical problem.

I do this work independently and vendor-neutral, so I'll say the plain version: a screen that's been handed between a crusher OEM, a media supplier, and the plant's own crew is exactly where a fault hides in the seams — everybody tuned their own part and nobody owned the separation. An independent second look tends to pay off most on that kind of handed-off fault, on this line especially where feed moisture, media choice, and machine motion all point fingers at each other.